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Individual Project

Introduction

The dataset our team chose is Dataset 5, which relates to NBA draft picks during the near decade long span between 2001 and 2010. We decided that our Y-variable would be "Points Rebounds Assists Per Game", while the team uses "Free throw Percentage", "Value Over Replacement Player", "Win Share Per 48 Minutes", and "3 Point Percentage" as X variables; of which this paper will explore only one, latter. 3 Point Percentage, also known as 3P%, is a statistic in basketball that measures how efficiently a team or in our case a player makes 3-point field goals. It is measured through the ratio of 3-point shots made from the total 3-point shots attempted. This statistic is used in basketball to evaluate shooting abilities of a player in this current meta of basketball where 3-point goals are the standard; the typical professional Basketball, or NBA (National Basketball Association), 3-Point Percentage is around 35-40%. I believe that most draft picks would be within the NBA standard for 3P% perhaps ranging on the lower side of 20-35%, I don't believe that it would have that strong of a correlation with "Points Rebounds Assists Per Game" (PRAPG) as someone with strong 3-point percentage would have less rebounds or assists.

$$3P\% = (\frac{3-point\ shots\ made}{3-point\ shots\ attempted}) \times 100$$

Equation 1: 3P% Formula

Describing the Data: Histogram

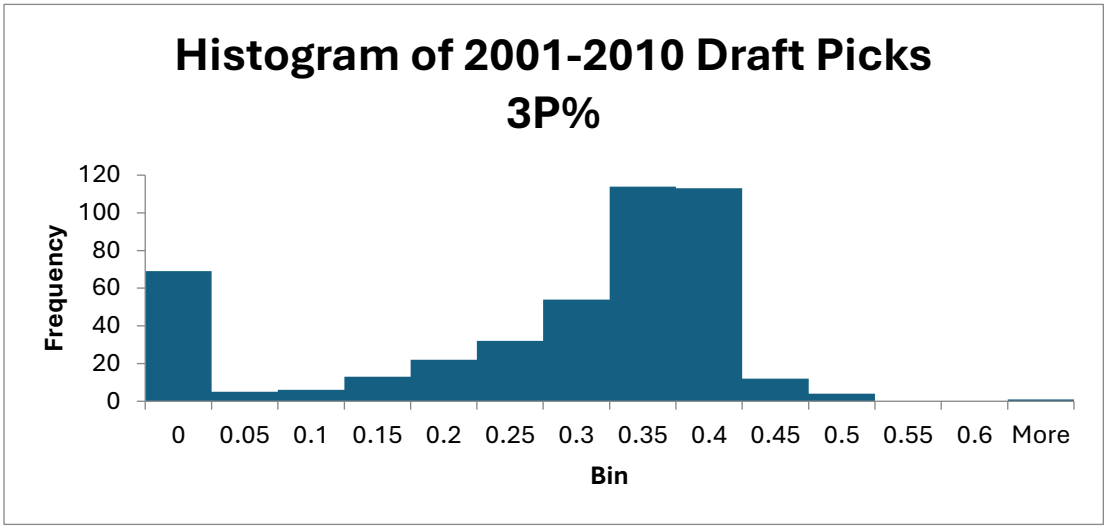


Figure 1: Histogram of 2001-2010 Draft Picks 3P%

When the X-variable (3P%) data is graphed on a histogram (Figure 1), using 5% bins, the data points all fall roughly between 0% and 54.9%, however there is a sharp drop off of 3P% after 45% which is proven by almost no players shooting above 50%. The distribution is skewed left as most of the data is within the higher percentage bins (0.3-0.4); the long tail on the left supports the negative skew with many players having very low 3P% with the 0.0-0.05 bin having the 3rd largest bin within the histogram. The highest frequency bin for this set is 0.35-0.40, indicating that the peak (mode) is within this range as this was the most common, recurring, section of numbers. The outlier of the 0 bin, being unusually large suggests that players attempt very few 3-point shots (maybe even 0). Overall, the shape of this graph shows that while some players are strong 3-point shooters, many others don't attempt 3-pointers or are poor in their execution; this could perhaps be due to changes in playstyle throughout the years.

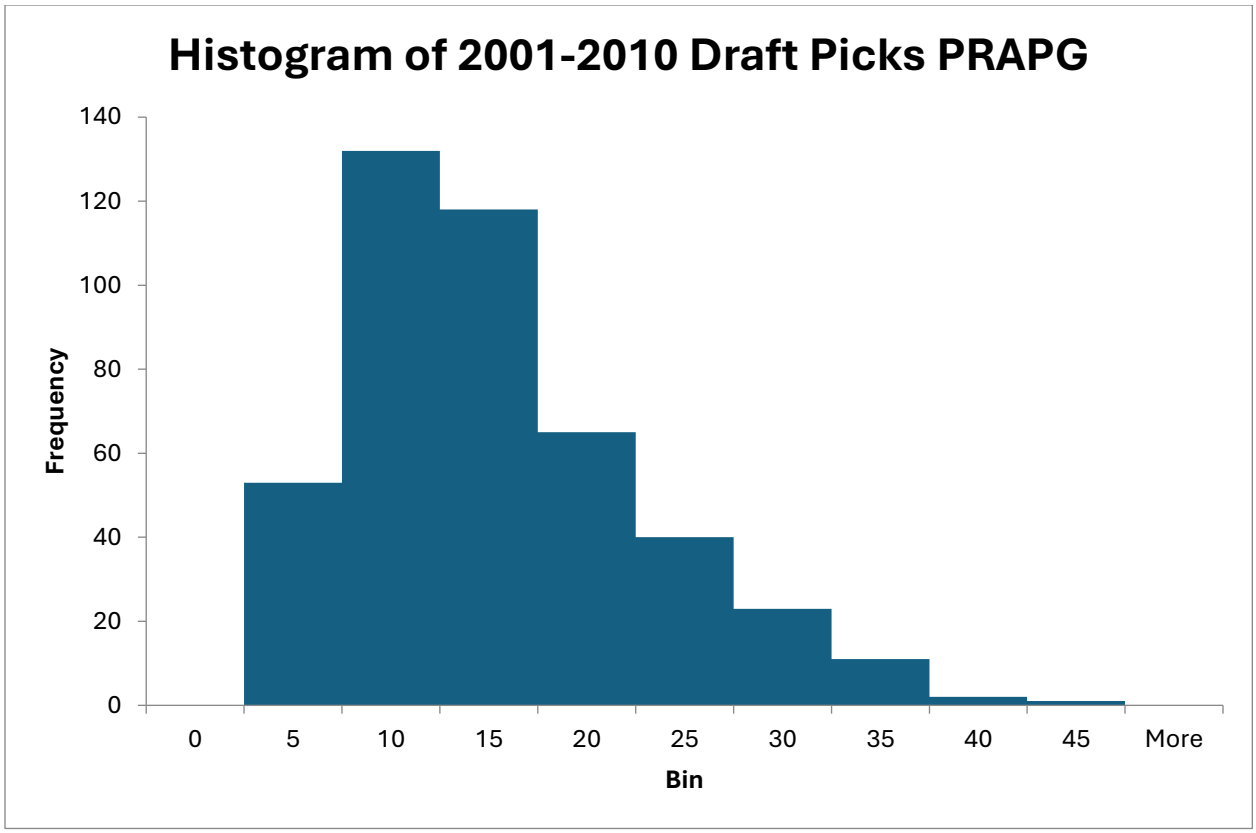


Figure 2: Histogram of 2001-2010 Draft Picks PRAPG

When looking at the Y-Variable histogram (Figure 2) of "Points, Rebounds, Assists Per Game" (PRAPG) using 5-point bins the data falls between 5 and 45 points. We see that its distribution is positively skewed towards the right with most players falling in the 5-20 PRAPG range there is a long tail towards the right, indicating that star players (with many points and assists) contribute with very high (35+) PRAPG totals. The highest section is 10-15 PRAPG sugessting that this is the most common PRAPG score across all players. There are no noticeable outliers within the graph but only a small number of playes exceed 30 PRAPG showing its rarity in frequency. There is also a noticable number of players in the 5-10 PRAPG which might be attributated to non-frequent players such as substitutes, injured, and benched players. In comparison to the X-axis graph it shows that most players have a moderate impact on PRAPG, even if some lack in 3P%.

Empirical Rule

When reviewing the 3P% data in excel we can look for the empirical rule to determine wether our data falls within the standard normal distribution. We can see (Table 1) that 303 datapoints fall within 1 standard deviation, 354 datapoints fall within 2 standard deviations, and 371 datapoints fall within 3 standard deviations. This equates to 68%, 80%, and 83% respectively when the datapoints are divided by the sample size, 445. This wholly means that the

3P% does not fall within the normal distribution, meaning that the graph is scewed, which we can see in the "outliers" within the 0-0.05 bins in Figure 1.

Mean	Standard Dev	Sample Size				
0.309	0.0887	445				
	-	+	count	% of 445	Rule %	
1 SD	0.2203	0.3977	303	0.680899	68	TRUE
2 SD	0.1316	0.4864	354	0.795506	95	FALSE
3 SD	0.0429	0.5751	371	0.833708	99.7	FALSE

Table 1: Empirical Rule Table for 3P%

Reviewing PRAPG however presents a different vision as this data seems to look more in line with a standard distribution. In Table 2 we see that 315 datapoints fall within 1 standard deviation, 424 datapoints fall within 2 standard deviations, and 443 datapoints fall within 3 standard deviations. Which more closeley equates to 70%, 95%, and 99.5% respectively falling fully in line with the normal standard distribution

Mean	Standard Dev	Sample Size				
12.96584	7.527122252	445				
	-	+	count	% of 445	Rule %	
1SD	5.438720445	20.49296	315	0.707865	68	TRUE
2SD	-2.08840181	28.02009	424	0.952809	95	TRUE
3SD	-9.61552406	35.54721	443	0.995506	99.7	TRUE

Table 2: Empirical Rule Table for PRAPG

Z-Scores and Outliers

To specifically find outliers within both datasets we have to establish Z-Scores for each datapoint within each dataset, we do this by following Equation 2, subtracting the mean from the value and dividing by the standard deviation. From here any z value above the absolute value of 3 is an outlier (within a standard distribution).

$$\frac{(x-\mu)}{\sigma}$$

Equation 2: Z-Score Formula

$$z > |3|$$

Equation 3: Outlier formula

To get the information we need we will need to run descriptive analysis using excel to ensure we get the most accurate data.

PRAPG	
Mean	12.96584
Standard Error	0.35682
Median	11.9
Mode	5.5
Standard Deviation	7.527122
Sample Variance	56.65757
Kurtosis	0.595537
Skewness	0.920262
Range	40.3
Minimum	1
Maximum	41.3
Sum	5769.8
Count	445

Table 3: Descriptive Statistics for PRAPG

3PP	
Mean	0.261067
Standard Error	0.006564
Median	0.314
Mode	0
Standard Deviation	0.138471
Sample Variance	0.019174
Kurtosis	1.213273
Skewness	-0.57728
Range	1
Minimum	0
Maximum	1
Sum	116.175
Count	445

Table 4: Descriptive Statistics for 3P%

Using the above information and formulas the following outliers can be determined

3P% X-Value Outlier(s)	PRAPG Y-Value Outlier(s)
1	41.3
	38.1

Table 5: List of Outliers

We can compile the information from the desctiftive summary intio a five number summary and later a box plot, this looks like the following:

3P%					
	Min	**Q1**	**Med**	**Q3**	**Max**
Score	0		0.314		1
Z-Score	-1.88536	-0.44101	0.382266	0.685579	5.336386583

Table 6 : 3P% Five Number Summary w/ Z-scores

PRAPG					
	Min	**Q1**	**Med**	**Q3**	**Max**
Score	1	7.2	11.9	16.9	41.3
Z-Score	-1.5897	-0.76601	-0.1416	0.522664	3.764275

Table 7: PRAPG Five Number Summary w/ Z-Scores

	3P%	Z-Scores	PRAPG	Z-Scores
Mean	0.261067	0	12.96584	0
Mode	0	-1.88536	5.5	-0.99186
Standard Dev.	0.138471	N/A	7.527	N/A

Table 8: Supplemental 5 table summary w/ Z-Scores

Regression

After using the excel data analysis add-in to complete regression analysis it generated the following data and plots (*residual and probability data too long, please request if needed):

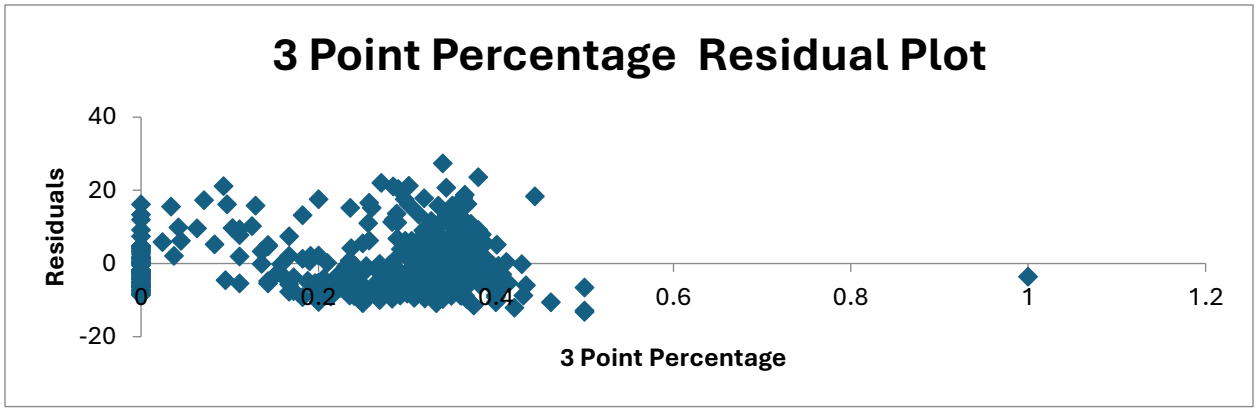


Figure 3: 3 Point Percentage Residual Plot

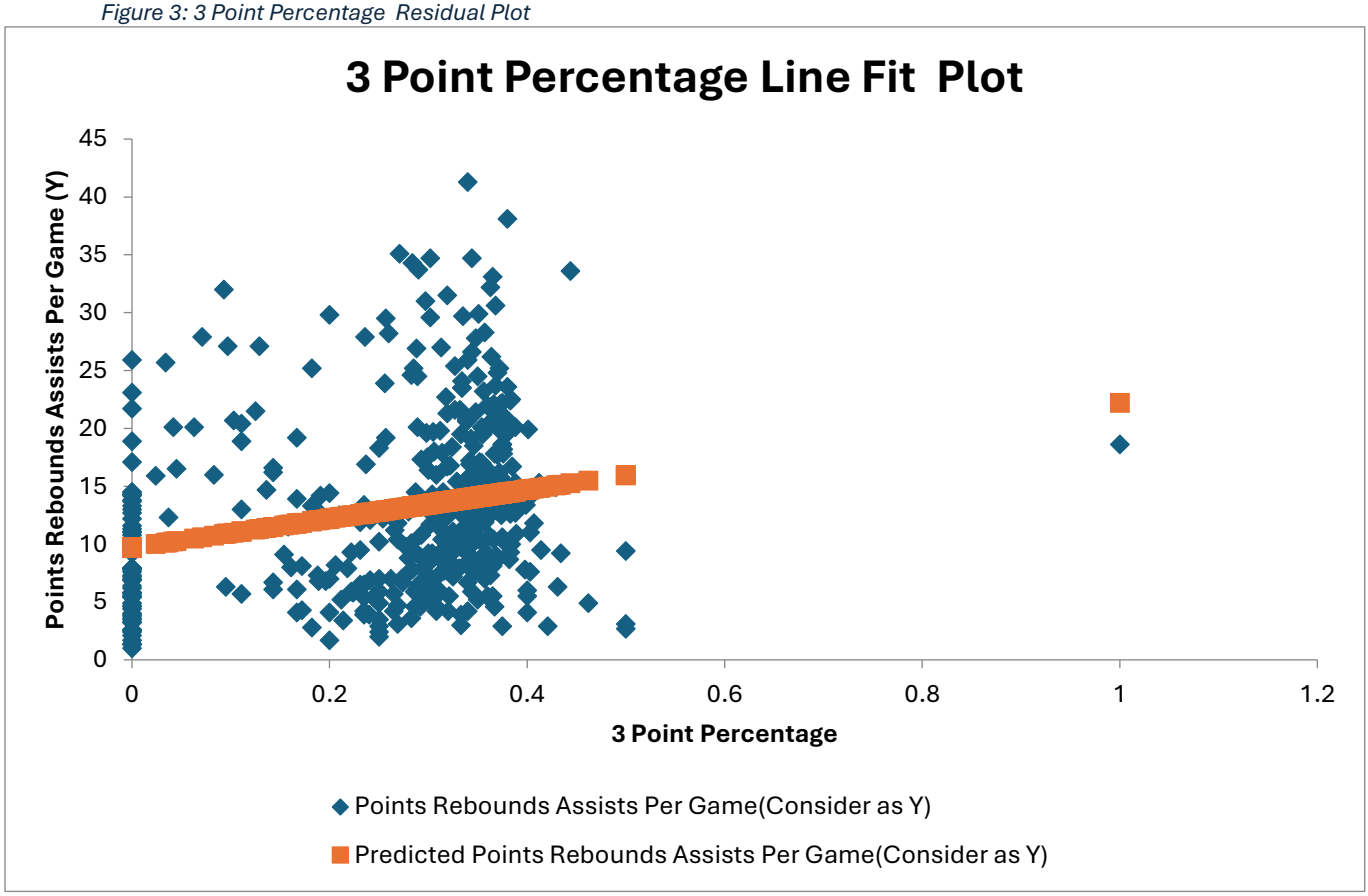


Figure 4: 3 Point Percentage Line Fit Plot

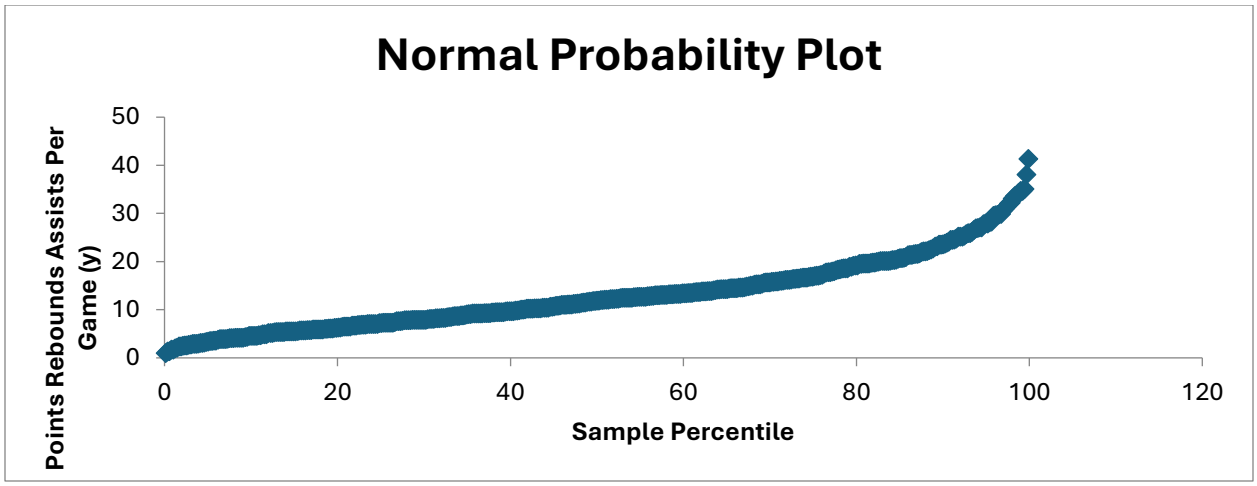


Figure 5: PRAPG & Sample Percentile Normal Probability plot

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.230075							
R Square	0.052934							
Adjusted R	0.050797							
Standard E	7.333455							
Observatio	445							
ANOVA								
	df	*SS*	*MS*	*F*	*ignificance F*			
Regression	1	1331.615	1331.615	24.76062	9.31E-07			
Residual	443	23824.35	53.77956					
Total	444	25155.96						
	Coefficients	*andard Erro*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*	*Upper 95.0%*
Intercept	9.700769	0.742566	13.06385	3.2E-33	8.24138	11.16016	8.24138	11.16016
3 Point Per	12.50663	2.513388	4.976004	9.31E-07	7.566985	17.44628	7.566985	17.44628

Table 9: Regression Results Table (3/5)

A way to confirm if this coorelation works outside of the computer regressing for us is through line fit, I've prepared the following scatter plot (Figure 6). From here we are able to somewhat observe that as x increases so does y (especialy between x=0.2-0.4), in essence as 3P% increases the PRAPG roughly increases along with it. It is hard to confirm by eye when looking at the randomness but this is confirmed when we look at the computer generated line fit plot (Figure 4) which shows a prediction line for PRAPG as 3P% increases. There is enough evidence to detirmine that there is a linear relationship between 3P% and PRAPG. As it seems 3-pointers prominence within scoring helps bring up many different factors within the PRAPG statistic.

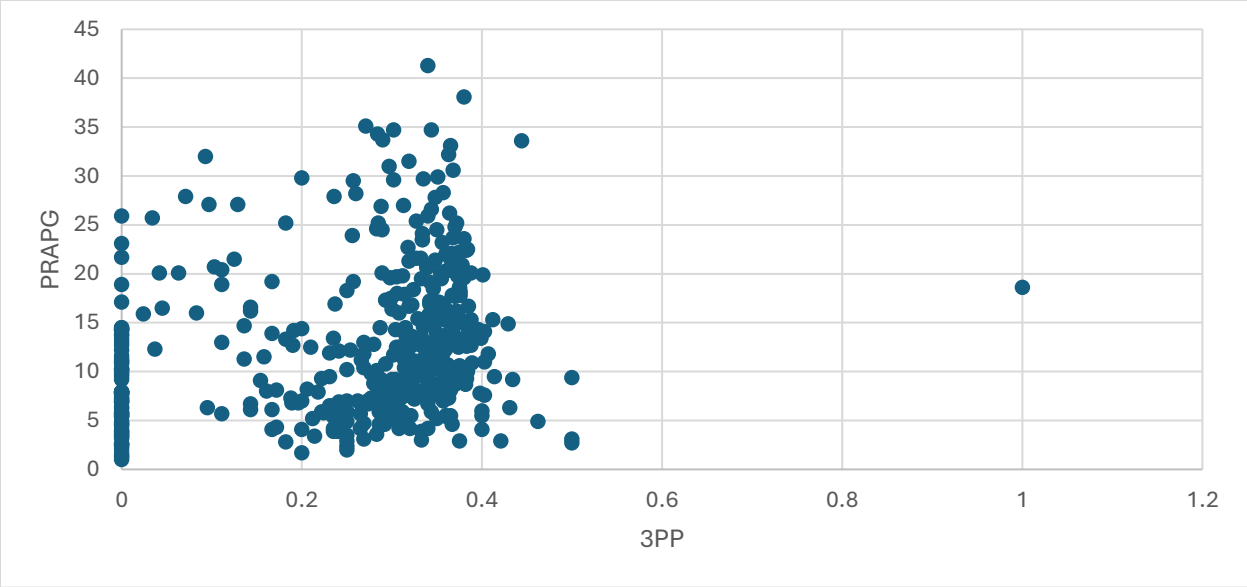


Figure 6: Scatter Plot

After we have determined a general relationship between the two variables we have to be able to prove its statistical significance, we can do this within the work we did in regression (Table 9) by testing p values. We check overall model significance by checking ANOVA and F-test values which we check for their significance values and whether they are <0.05, because if they are they contribute significantly to the model (at 95% confidence). If we also test p-values of the t-stats table to see if its <0.05 we can also determine if it's a significant contributor. We can also check R^2 and adjusted R^2 values if they are >0.5 then that means it is a strong fit. According to our data (Table 9) we heave a pretty good fit, although it is very close to 0.5. We can also check our residual plots (Figure 3) to check if there are any unusual patterns which occur maybe due to an error in data. Ours seem random enough meaning we can be sure that our data is clean. According to the above methods there is a significance between the two variables. While it is significant this single x variable is not a reliable predictor, even logically thinking a lot more go into the PRAPG statistic than just 3-pointer shots, this explains much of the variance within the data (Figure 6), despite the upward trend, different scoring percentages, rebound percentage, assist stat, and games played effect the correlation between 3P% and PRAPG.

Assumption Check:

As somewhat mentioned above some dangerous behaviours show up in our models to make it not a good predictor between x and y. The residuals plot (Figure 5) shows noticable curves making it not a linear coorelation and constant variance is off by a fan shaped plots (Figure 6).

Summary:

The established link between 3P% and PRAPG is a weak coorelation as predicted within the introduction. As 3-pointer consistency is not that strong of an indicatior of rebounds or assists per game. Through the above statistical models it has been demonstrated that more variables are needed to equate to PRAPG, a reduction to just Points per game (PPG) would contribute to more statistically clean model.

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